



DEALER BEST PRACTICE SERIES

Component Repair /
Rebuild / Reman Strategy

Application Maintenance Site Management **Component Rebuild** Component Life Management Safety MARC Management

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1.0 Introduction

This Best Practice is to communicate the key points of the Global Mining / Reman “Major Component Repair / Rebuild / Reman” strategy and to share a process used by several dealers to develop a CRC policy for the use of Reman components and subcomponents to support customer component rebuild requirements.

2.0 Best Practice Description

In 1999, a series of studies and projects were initiated to determine the best method to deliver Mining customer component rebuild requirements at the lowest cost per hour. It was clear that dealers with an efficient CRC and good salvage capabilities could deliver the lowest rebuild costs for 1st and 2nd life cycle component rebuilds. The ability to deliver economical rebuilds beyond the 2nd life cycle was dependant on the dealer capability and cost to salvage major castings, such as engine blocks and spindles. It was also clear that not all dealers had a CRC or advanced salvage capability.

A strategy was created with the following key points:

- Dealer rebuild for the 1st and 2nd component life cycles is the primary path to support Mining customer needs and deliver lowest cost per hour.
- For 3rd or higher life cycle rebuilds, or when dealer rebuilds are no longer economically practical, Reman major components become the primary path.
- Reman major components can support CRC workload cycles with peak shaving.
- Reman minor subcomponents can support CRC rebuilds.
- Reman major component pricing was positioned to enable / encourage dealer CRC investment.
- For dealers with low machine populations and limited CRC capability, the Reman option for major components serve as the primary path to support Mining customer needs.

Prior to 2003, Reman major component pricing was generally based on a “percentage of new” formula. In 2003, new pricing formulas were established based on “Dealer Centerline” 2nd rebuild parts and labor price. Reman prices were to be set at a premium above average dealer 2nd rebuild price. The “premium” was set at 10% for component rebuilds priced over \$60,000 and higher premiums (10%-45%) for lower cost components.

“Dealer Centerline 2nd Rebuild Parts and Labor Price” data is determined from Repair Option Builder. Repair option costs are adjusted year to year per dealer input.

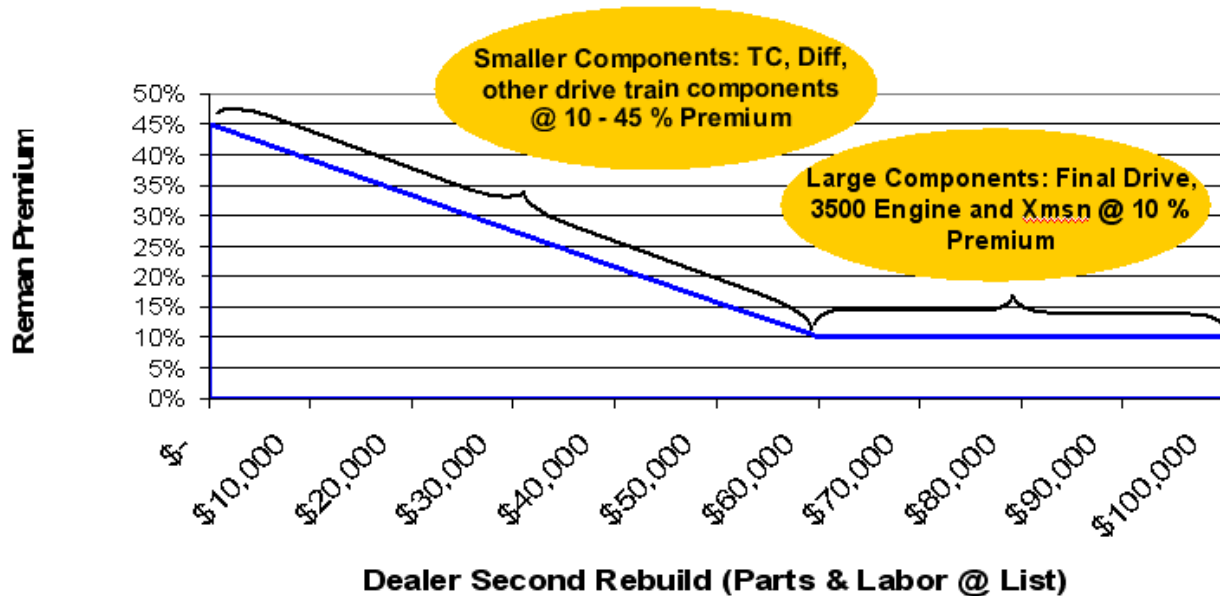
This pricing formula was established to assure that an efficient CRC could complete with Reman prices and justify CRC investments.

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See chart for target premiums:



Current Reman Major Component Pricing Strategy



Component Rebuild Cost Data

Prior to 2003 / 2004, 2nd component rebuild cost data contained in “Repair Option Builder” was determined with a limited level of validation against actual dealer cost information. Today, that process has changed. Data today is based on an analysis of a large number of actual dealer work orders. Annual conferences are held with a number of key dealerships to fine tune the data, document actual parts reuse and salvage costs, and to identify the cost drivers and the cost of required updates. The centerline price data today, especially for the larger mining machines, reflects “real world” dealer costs and prices.

Still, some dealer’s rebuild costs are higher than the centerline. Centerline data is by definition a world average rebuild price. From one dealer to another, there is often a level of variation due to many factors, such as labor rate or salvage capability, etc.

The chart below illustrates the variation in rebuild costs from one dealer to another. In the chart, dealer “A” has low labor rates and good parts salvage capability, dealer “B” has much higher labor rates, Dealer “C” has limited machine shop / salvage capability. The key point is that not all dealer CRC average component rebuild costs are the same and may be either above or below the world average.

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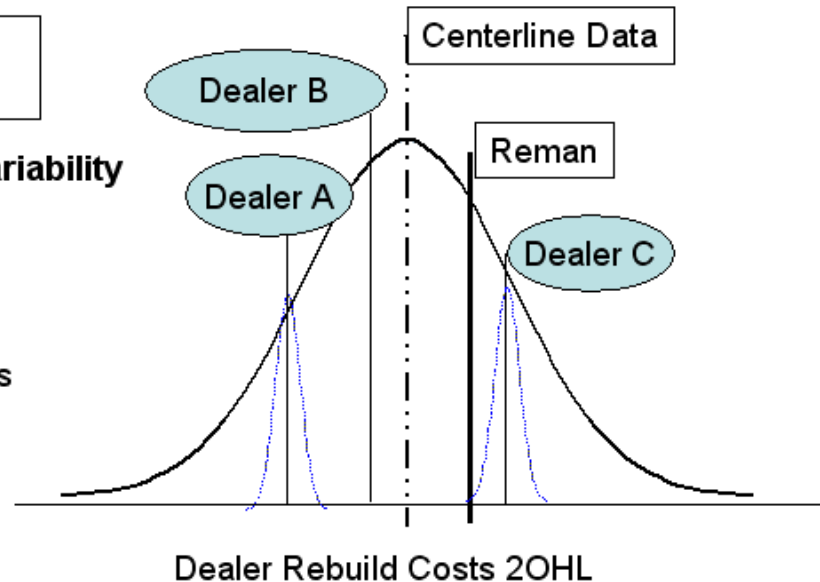
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Between Dealer Variability
Much greater than With-in

Causes of Between Dealer Variability

1. Local labor rates
2. Salvage / reuse practices
3. Local duties & taxes
4. Regional pricing
5. FDO discounts / MPC plans



Within the strategy, there are three major roles that Reman components and subcomponents play in the support of a CRC. Those are:

- Peak shaving when CRC workloads exceed capacity and / or inventories of exchange components are low.
- Using Reman subcomponents in rebuilds to accelerate the rebuild or when dealer salvage costs exceed the cost of a Reman part or subcomponent.
- Replacing high hour major components when rebuild and salvage costs are expected to exceed Reman cost.

Many dealers have developed a formal internal policy to guide decisions on when to use Reman and when to rebuild. The policies are based on rebuild cost and process analysis. Developing a formal CRC Reman usage policy guides decisions and lowers CRC costs.

3.0 Implementation

The following are the key considerations for creating a CRC policy for using Reman products.

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Peak Shaving Policy

Most mining customers use exchange components to maximize machine availability. The inventory of exchange components may either be dealer owned or customer owned components. This inventory should have a “minimum stock” level of inventory defined by machine model and by component. “Minimum” stock is usually equal to a 30-60 day demand.

If exchange component inventory falls below the minimum stock level and the CRC cannot supply needed demand, Reman major components should be considered to manage the situation until exchange inventories can be replenished.

Considerations include:

- Current and scheduled CRC workload.
- CRC Component Turnaround Time
- Monthly exchange inventory demand and minimum stock level.
- Time required for CRC to replenish exchange component inventory to above minimum, given current workload.
- Other options, such as exchange inventory seed with new components.
- Time required to order and receive a Reman component.

The ability to plan and schedule incoming rebuilds through good customer communications is vital to the process. If incoming component rebuilds can be planned several months in advance, workload peaks are more manageable.

An example of a Peak Shaving Policy is:

“If exchange inventory of 793C engines falls below 10 units (30 day normal demand) and the CRC cannot replenish the inventory within 10 days, a Reman engine is ordered”.

Use of Reman Parts and Subcomponents

Most dealer CRCs have a standard parts rebuild kit to define the parts, including Reman subcomponents, which are to be used in the rebuild. The kit also defines which components will be rebuilt / salvaged by the CRC or locally outsourced. The decisions are based on a dealer rebuild cost versus Reman cost analysis.

As example, Reman offers 3500 series engine cylinder head groups, bare cylinder head, and upgrade to new. A dealer cost review would determine the parts, labor, and salvage cost to rebuild a cylinder head group and a bare cylinder head. Those costs would be compared to Reman costs. Depending on labor rates, labor hours, dealer salvage ability and capacity, the dealer would select the most economical option. The best option then becomes the CRC standard practice. Dealer may choose to standardize on using Reman Cylinder Head groups, or to do their own rebuilds.

Other considerations should include the impact on component turnaround time. In the example above, cylinder heads can be rebuilt at the same time that other work is being done on the engine and probably does not cause a delay. Salvaging a spindle for a final drive may, however, add several days to the component rebuild turnaround time. The delay may cost more than it is worth and Reman may be a better choice to accelerate the rebuild.

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Another consideration may be warranty coverage and cost or current product updates / improvements.

Subcomponent Policy example: 3500 Engine Rebuilds

Reman subcomponents in standard rebuild kits: Always use.

- Water pumps (always use)
- Alternators (always use)
- Oil Coolers (always use)
- Turbo cartridges (always use)
- Cylinder Pack (always use)
- Injectors (always use)

Reman subcomponents for use if core requires salvage.

- Crankshaft
- Cylinder Head Group
- Camshaft

High Hour Major Components

When a dealer rebuild becomes more expensive than Reman, or when the level of salvage required exceeds dealer capability, Reman major components should be considered. Some dealers have a clear policy such as 1st and 2nd rebuilds are done in the CRC, 3rd rebuilds are replaced with Reman.

Some dealers as a standard practice will do the 3rd rebuild and use Reman thereafter.

The ability and cost to salvage major castings is the determining factor to consider.

Another consideration for very high hour parts is fatigue life and the component life expected.

Each dealer needs to determine what is the “economical” rebuild life cycle for the territory.

Economical rebuild life cycles can be different by machine model, by customer, and by component.

The cost must also be balanced against the component life expectations. Typically, 3rd life cycle or higher rebuilds on a component may not deliver the same life as previous rebuilds.

Specific policies can be created based on high hour rebuild cost history and component life data.

One important requirement is to be able to track component life cycles and hours. The hours and rebuild history information needs to be available when the component arrives to the shop. The decision to rebuild or replace with a Reman component should be made before work is started on the component. This means that the CRC must have easy access to component history and life information.

The same can be said for tracking individual key parts. Reused key parts (such as gears, connecting rods, etc) should be marked when reused with the hours accumulated.

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4.0 Benefits

- Lower CRC rebuild costs.
- Improve CRC and MARC margins.
- Fully utilize CRC assets
- Lower customer cost per hour.
- Improve CRC turnaround time.

5.0 Resources Required

- Team to develop dealer policies.
- A component life tracking and rebuild history database.
- A CRC planning and scheduling process.
- Standard jobs and rebuild kits.

6.0 Supporting Attachments / References

7.0 Related Best Practices

1006.42.1030 Improving CRC Turnaround Time Through Machine Shop Subcomponent Exchange Program

8.0 Acknowledgements

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